

G. Narayanamma Institute of Technology & Science

(Autonomous) (for Women)
Shaikpet, Hyderabad- 500 104

IV-B.Tech I-Semester *Regular Examinations, November - 2025*

DATA SCIENCE USING R (Computer Science and Engineering)

Max. Marks: 60

Time: 03 Hours

Note:

1. Question paper comprises of **Part A** and **Part B**.
2. **Part A** (10 marks) is **compulsory** and consists of **ten sub-questions**, from **all units**, each carrying **1 mark**.
3. **Part B** (for 50 marks) consists of **five questions** with **“either” “or”** pattern. Each question carries 10 marks and may have a,b,c as sub questions. The student has to answer any one full question.

PART-A

(Answer 10 questions. Each question carries 1 mark)

Q.No.	Question	Marks	CO	BTL
Q.1	a) Write any two advantages of using R for data exploration.	[01]	CO1	[L1]
	b) What are the main stages of a data science project?	[01]	CO1	[L1]
	c) Define a Data Frame in R.	[01]	CO2	[L1]
	d) Mention one method to handle missing values in data.	[01]	CO2	[L2]
	e) Differentiate between single-variable and multivariable models.	[01]	CO3	[L3]
	f) Why training data and validation data different?	[01]	CO6	[L1]
	g) State one real-world application of k-means clustering.	[01]	CO4	[L2]
	h) Explain the purpose of cluster analysis in machine learning.	[01]	CO4	[L2]
	i) Define time series data.	[01]	CO5	[L1]
	j) Apply ARIMA models in any one real-world application.	[01]	CO5	[L3]

PART-B

(Answer 05 full questions. Each question carries 10 marks)

Q.No.	Question	Marks	CO	BTL
Q.2(a)	Explain the stages of a data science project with a suitable example.	[05]	CO1	[L2]
(b)	Describe the steps involved in installing and setting up R and R Studio.	[05]	CO1	[L4]
OR				
Q.3(a)	Discuss different data types in R with examples.	[05]	CO1	[L3]
(b)	Explain the process of loading and handling data in R using files and databases.	[05]	CO1	[L2]

Q.4(a)	Write R commands to load a CSV file and display its first 10 rows.	[05]	CO2	[L1]
(b)	Explain different methods to handle missing values in R with examples.	[05]	CO2	[L4]
OR				
Q.5(a)	Explain how data visualization helps in spotting problems in data.	[05]	CO2	[L2]
(b)	Describe the steps for cleaning and preparing data for modeling and validation.	[05]	CO2	[L3]
Q.6(a)	Explain the process of choosing, evaluating, and validating models in data science.	[05]	CO3	[L2]
(b)	Describe the KDD process with reference to knowledge discovery from data.	[05]	CO3	[L3]
OR				
Q.7(a)	Discuss the importance of model validation and different validation techniques.	[05]	CO3	[L1]
(b)	Explain the difference between single-variable and multi-variable models with examples.	[05]	CO6	[L3]
Q.8(a)	Describe logistic regression and its advantages over linear regression.	[05]	CO4	[L2]
(b)	Explain the difference between supervised and unsupervised methods with examples.	[05]	CO6	[L4]
OR				
Q.9(a)	Discuss the steps of k-means clustering with an example.	[05]	CO4	[L2]
(b)	Describe the process of building and interpreting a logistic regression model in R.	[05]	CO4	[L3]
Q.10(a)	Explain the components of time series data and their significance.	[05]	CO5	[L2]
(b)	Describe how time series data can be decomposed using R.	[05]	CO5	[L4]
OR				
Q.11(a)	Discuss ARIMA models for time series forecasting.	[05]	CO5	[L1]
(b)	Explain the process of plotting and analyzing time series data in R.	[05]	CO5	[L3]

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